

4.25

(a)

```
=====
                        值
-----
slope      9.020
elasticity 0.983
-----
```

- 當房屋面積 (sqft) 增加 1 單位時，房價 (price) 預計平均增加 9.020 單位
- 當房屋面積 (sqft) 增加 1% 時，房價 (price) 預計平均增加約 0.983%

(b)

```
=====
                        值
-----
slope      9.400
elasticity 1.025
-----
```

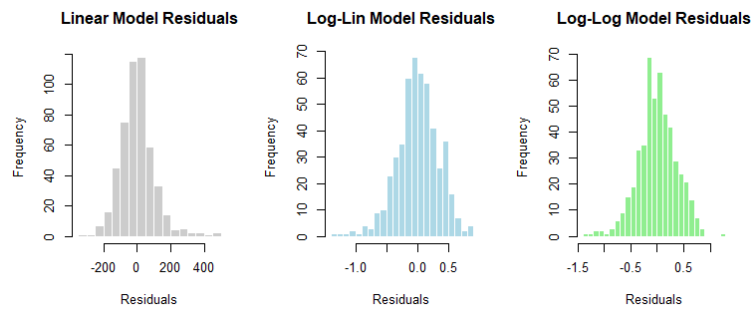
- 當房屋面積 (sqft) 增加 1% 時，房價 (price) 預計平均增加 1.025%
- 當房屋面積 (sqft) 增加 1 單位時，房價 (price) 預計平均增加 9.400 單位

(c)

```
=====
R_squared_Type value
-----
Linear Model Standard R2  0.641
Log-Lin (a) Generalized R2 0.662
Log-Log (b) Generalized R2 0.645
-----
```

- 若 Log-Lin 模型的 Generalized R2 最高，則代表該模型對房價實際數值的解釋能力最強

(d)



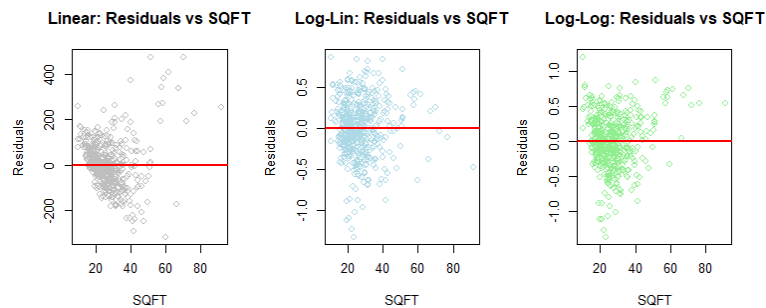
=====

JB_Stat p_value

Linear	221.112	0
Log-Lin	26.679	0.00000
Log-Log	14.279	0.001

- 三個模型的 JB 檢定 p-value 均小於 0.05，在統計學上皆拒絕了殘差為常態分佈的虛無假設

(e)



- 線性模型呈現明顯的異質變異性，而對數模型（尤其是 Log-Log）有效修正了此問題，使殘差分佈更趨於同質變異且隨機

(f)

=====	
Model	Predicted_Price

1 Linear	246.456
2 Log-Lin	214.234
3 Log-Log	227.539

(g)

=====			
Model	Fit	Lower	Upper

1 Linear	246.456	44.277	448.634
2 Log-Lin	214.234	109.768	418.117
3 Log-Log	227.539	111.141	465.841

- 基於 Log-Log 模型 在修正異質變異性、提升 Generalized R2 以及殘差常態性表現上均為最優，我們認為該模型是描述 collegetown 房價與面積關係的最佳函數形式

5.20

(e)

=====				
	N.100	N.500	N.1000	N.5000

Mean_var_ols	1.225	0.241	0.120	0.024
Mean_var_mean	1.652	0.322	0.161	0.032
E[1/sx2]	0.122	0.120	0.120	0.120
Ex_var	0.165	0.161	0.161	0.160

(f)

The simulation confirms that while both estimators are consistent, the OLS estimator is more efficient due to its lower variance, with $E[(s_x^2)^{-1}]$ and $E[\frac{4}{(\bar{x}_2 - \bar{x}_1)^2}]$ converging to 0.12 and 0.16, respectively.

4.1 Answer each of the following:

- a. Suppose that a simple regression has quantities $N = 20$, $\sum y_i^2 = 7825.94$, $\bar{y} = 19.21$, and $SSR = 375.47$, find R^2 .
- b. Suppose that a simple regression has quantities $R^2 = 0.7911$, $SST = 725.94$, and $N = 20$, find $\hat{\sigma}^2$.
- c. Suppose that a simple regression has quantities $\sum (y_i - \bar{y})^2 = 631.63$ and $\sum \hat{e}_i^2 = 182.85$, find R^2 .

$$\begin{aligned} a. \quad R^2 &= \frac{SSR}{SST} = \frac{375.47}{7825.94 - 20 \cdot 19.21^2} \\ &= 0.8429 \end{aligned}$$

$$\begin{aligned} b. \quad R^2 &= \frac{SSR}{SST} = 1 - \frac{SSE}{SST} \\ SSE &= SST(1 - R^2) = 151.6489 \\ MSE &= \frac{SSE}{20-2} = 8.4249 = \hat{\sigma}^2 \end{aligned}$$

$$c. \quad R^2 = \frac{631.63 - 182.85}{631.63} = 0.7105$$

4.3 We have five observations on x and y . They are $x_i = 3, 2, 1, -1, 0$ with corresponding y values $y_i = 4, 2, 3, 1, 0$. The fitted least squares line is $\hat{y}_i = 1.2 + 0.8x_i$, the sum of squared least squares residuals is $\sum_{i=1}^5 \hat{e}_i^2 = 3.6$, $\sum_{i=1}^5 (x_i - \bar{x})^2 = 10$, and $\sum_{i=1}^5 (y_i - \bar{y})^2 = 10$. Carry out this exercise with a hand calculator. Compute

- the predicted value of y for $x_0 = 4$.
- the $se(f)$ corresponding to part (a).
- a 95% prediction interval for y given $x_0 = 4$.
- a 99% prediction interval for y given $x_0 = 4$.
- a 95% prediction interval for y given $x = \bar{x}$. Compare the width of this interval to the one computed in part (c).

a. $\hat{y}_0 = 1.2 + 0.8 \cdot 4 = 4.4$

b. $MSE = \frac{3.6}{5-2} = 1.2$, $\bar{x} = 1$

$$se(f) = \sqrt{Var(f)} = \sqrt{1.2 \cdot \left[1 + \frac{1}{5} + \frac{(4-1)^2}{10} \right]} = 1.5875$$

c. $\hat{y}_0 \pm t_{0.05}(5-2) se(f)$

$$= [4.4 \pm 3.182 \cdot 1.5875] = [-0.614, 9.454]$$

d. $[4.4 \pm 5.841 \cdot 1.5875] = [-4.8726, 13.6726]$

e. $se(f) = \sqrt{1.2 \cdot \left(1 + \frac{1}{5} \right)} = 1.2$

$$\bar{x} = 1$$

$$[1 \pm 3.182 \cdot 1.2] = [-1.8184, 5.8184]$$

→ 當 $x_0 = \bar{x}$, P.I. 最窄, 隨著離 $\bar{x}$ 愈遠愈寬。

5.5 For each of the following two time-series regression models, and assuming MR1–MR6 hold, find $\text{var}(b_2|\mathbf{x})$ and examine whether the least squares estimator is consistent by checking whether $\lim_{T \rightarrow \infty} \text{var}(b_2|\mathbf{x}) = 0$.

- a. $y_t = \beta_1 + \beta_2 t + e_t$, $t = 1, 2, \dots, T$. Note that $\mathbf{x} = (1, 2, \dots, T)$, $\sum_{t=1}^T (t - \bar{t})^2 = \sum_{t=1}^T t^2 - \left(\sum_{t=1}^T t\right)^2 / T$, $\sum_{t=1}^T t = T(T+1)/2$ and $\sum_{t=1}^T t^2 = T(T+1)(2T+1)/6$.
- b. $y_t = \beta_1 + \beta_2 (0.5)^t + e_t$, $t = 1, 2, \dots, T$. Here, $\mathbf{x} = (0.5, 0.5^2, \dots, 0.5^T)$. Note that the sum of a geometric progression with first term r and common ratio r is

$$S = r + r^2 + r^3 + \dots + r^n = \frac{r(1 - r^n)}{1 - r}$$

- c. Provide an intuitive explanation for these results.

$$a. \text{Var}(b_2|\mathbf{x}) = \frac{\sigma^2}{\sum_{t=1}^T (x_t - \bar{x})^2}$$

$$\begin{aligned} x_t &= t, \quad \sum_{t=1}^T (x_t - \bar{x})^2 = \sum_{t=1}^T (t - \bar{t})^2 = \sum_{t=1}^T t^2 - \frac{(\sum_{t=1}^T t)^2}{T} \\ &= \frac{T(T+1)(2T+1)}{6} - \frac{\left[\frac{T(T+1)}{2}\right]^2}{T} \\ &= \frac{T^3 - T}{12} \end{aligned}$$

$$\lim_{T \rightarrow \infty} \text{Var}(b_2|\mathbf{x}) = \lim_{T \rightarrow \infty} \frac{12\sigma^2}{T^3 - T} = 0, \quad \text{consistent}$$

$$b. \sum_{t=1}^T x_t = 0.5 + 0.5^2 + \dots + 0.5^T = \frac{0.5(1 - 0.5^T)}{1 - 0.5} = 1 - 0.5^T$$

$$\sum_{t=1}^T x_t^2 = 0.5^2 + (0.5^2)^2 + \dots + (0.5^T)^2 = \frac{0.5^2(1 - 0.5^{2T})}{1 - 0.5^2} = \frac{1 - 0.5^{2T}}{3}$$

$$\lim_{T \rightarrow \infty} \sum_{t=1}^T (x_t - \bar{x})^2 = \lim_{T \rightarrow \infty} \frac{1 - 0.5^{2T}}{3} - \frac{(1 - 0.5^T)^2}{T} = \frac{1}{3}$$

$$\lim_{T \rightarrow \infty} \text{Var}(b_2|\mathbf{x}) = \lim_{T \rightarrow \infty} \frac{\sigma^2}{\frac{1}{3}} = 3\sigma^2 \neq 0, \quad \text{not consistent}$$

5.6 Suppose that, from a sample of 63 observations, the least squares estimates and the corresponding estimated covariance matrix are given by

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} \quad \widehat{\text{cov}}(b_1, b_2, b_3) = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

Using a 5% significance level, and an alternative hypothesis that the equality does not hold, test each of the following null hypotheses:

- a. $\beta_2 = 0$
- b. $\beta_1 + 2\beta_2 = 5$
- c. $\beta_1 - \beta_2 + \beta_3 = 4$

a. ① $\begin{cases} H_0: \beta_2 = 0 \\ H_a: \beta_2 \neq 0 \end{cases}$ ② $\alpha = 0.05$ ③ $T = \frac{b_2 - 0}{SE(b_2)} \sim t(60)$

④ $RR = \{ |T| \geq t_{0.025}(60) \Rightarrow \}$ ⑤ $T_0 = \frac{3}{\sqrt{4}} = 1.5$

⑥ $T_0 \notin RR$, do not reject H_0

b. ① $\begin{cases} H_0: \beta_1 + 2\beta_2 = 5 \\ H_a: \beta_1 + 2\beta_2 \neq 5 \end{cases}$ ② $\alpha = 0.05$ ③ $T = \frac{b_1 + 2b_2 - 5}{SE(b_1 + 2b_2)} \sim t(60)$

④ $RR = \{ |T| \geq t_{0.025}(60) \Rightarrow \}$ ⑤ $SE = \sqrt{3 + 4 \cdot 4 + 4 \cdot (-2)} = \sqrt{11}$
 $T_0 = \frac{2 + 2 \cdot 3 - 5}{\sqrt{11}} = 0.9045$

⑥ $T_0 \notin RR$, do not reject H_0

c. ① $\begin{cases} H_0: \beta_1 - \beta_2 + \beta_3 = 4 \\ H_a: \beta_1 - \beta_2 + \beta_3 \neq 4 \end{cases}$ ② $\alpha = 0.05$ ③ $T_0 = \frac{b_1 - b_2 + b_3 - 4}{SE(b_1 - b_2 + b_3)} \sim t(60)$

④ $RR = \{ |T| \geq t_{0.025}(60) \Rightarrow \}$ ⑤ $SE = \sqrt{3 + 4 + 3 - 2 \cdot (-2) - 2(0) + 2 \cdot 1} = 4$
 $T_0 = \frac{2 - 3 - 1 - 4}{4} = -1.5$

⑥ $T_0 \notin RR$, do not reject H_0

5.20 A generalized version of the estimator for β_2 proposed in Exercise 2.9 by Professor I.M. Mean for the regression model $y_i = \beta_1 + \beta_2 x_i + e_i, i = 1, 2, \dots, N$ is

$$\hat{\beta}_{2,mean} = \frac{\bar{y}_2 - \bar{y}_1}{\bar{x}_2 - \bar{x}_1}$$

where $(\bar{y}_1, \bar{x}_1)$ and $(\bar{y}_2, \bar{x}_2)$ are the sample means for the first and second halves of the sample observations, respectively, after ordering the observations according to increasing values of x . Given that assumptions MR1–MR6 hold:

- Show that $\hat{\beta}_{2,mean}$ is unbiased.
- Derive an expression for $\text{var}(\hat{\beta}_{2,mean} | \mathbf{x})$.
- Write down an expression for $\text{var}(\hat{\beta}_{2,mean})$.
- Under what conditions will $\hat{\beta}_{2,mean}$ be a consistent estimator for β_2 ?
- Randomly generate observations on x from a uniform distribution on the interval $(0, 10)$ for sample sizes $N = 100, 500, 1000$, and, if your software permits, $N = 5000$. Assuming $\sigma^2 = 1000$, for each sample size, compute:
 - $\text{var}(b_2 | \mathbf{x})$ and $\text{var}(\hat{\beta}_{2,mean} | \mathbf{x})$ where b_2 is the OLS estimator.
 - Estimates for $E[(s_x^2)^{-1}]$ and $E[4/(\bar{x}_2 - \bar{x}_1)^2]$ where s_x^2 is the sample standard deviation for x using N as a divisor.
- Comment on the relative magnitudes of your answers in part (e), (i) and (ii) and how they change as sample size increases. Does it appear that $\hat{\beta}_{2,mean}$ is consistent?
- Show that $E[(s_x^2)^{-1}] \xrightarrow{p} 0.12$ and $E[4/(\bar{x}_2 - \bar{x}_1)^2] \xrightarrow{p} 0.16$. [Hint: The variance of a uniform random variable defined on the interval (a, b) is $(b - a)^2/12$.]
- Suppose that the observations on x were not ordered according to increasing magnitude but were randomly assigned to any position. Would the estimator $\hat{\beta}_{2,mean}$ be consistent? Why or why not?

$$\begin{aligned} \text{a. } \bar{y}_1 &= \beta_1 + \beta_2 \bar{x}_1 + \bar{e}_1 \\ \bar{y}_2 &= \beta_1 + \beta_2 \bar{x}_2 + \bar{e}_2 \\ E(\hat{\beta}_{2,mean}) &= \frac{(\beta_1 + \beta_2 \bar{x}_2 + \bar{e}_2) - (\beta_1 + \beta_2 \bar{x}_1 + \bar{e}_1)}{\bar{x}_2 - \bar{x}_1} \\ &= \beta_2 + E\left(\frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1} | \mathbf{x}\right) = \beta_2 \end{aligned}$$

$$\text{b. } \text{Var}\left(\beta_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1} | \mathbf{x}\right) = \frac{1}{(\bar{x}_2 - \bar{x}_1)^2} \text{Var}(\bar{e}_2 - \bar{e}_1 | \mathbf{x})$$

$$\text{Var}(\bar{e}_1 | \mathbf{x}) = \text{Var}(\bar{e}_2 | \mathbf{x}) = \frac{\sigma^2}{\frac{N}{2}} = \frac{2\sigma^2}{N}$$

$$\text{Var}(\bar{e}_2 - \bar{e}_1 | \mathbf{x}) = \frac{4\sigma^2}{N} \Rightarrow \text{Var}(\hat{\beta}_{2,mean} | \mathbf{x}) = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}$$

$$\begin{aligned}
 c. \text{Var}(\hat{\beta}_{2, \text{mean}}) &= E(\text{Var}(\hat{\beta}_{2, \text{mean}}|X)) + \underbrace{\text{Var}(E(\hat{\beta}_{2, \text{mean}}|X))}_{\substack{\beta_2 \\ 0}} \\
 &= E(\text{Var}(\hat{\beta}_{2, \text{mean}}|X))
 \end{aligned}$$

$$d. \text{When } N \rightarrow \infty, \text{Var}(\hat{\beta}_{2, \text{mean}}|X) = \frac{4\sigma^2}{N(X_2 - X_1)^2} \rightarrow 0$$

$\rightarrow$ consistent

$$g. \sigma_X^2 = \frac{(10-0)^2}{12} = \frac{100}{12}$$

$$P[E[(X^2)^{-1}] \rightarrow \frac{1}{\sigma_X^2} = \frac{12}{100} = 0.12$$

$$\textcircled{a} X_1, X_2, \dots, X_N \sim U(0, 5) \rightarrow \bar{X}_1 = 2.5$$

$$X_{N+1}, \dots, X_N \sim U(5, 10) \rightarrow \bar{X}_2 = 7.5$$

$$\Rightarrow \bar{X}_2 - \bar{X}_1 = 5$$

$$E\left[\frac{4}{(\bar{X}_2 - \bar{X}_1)^2}\right] \xrightarrow{P} \frac{4}{5^2} = 0.16$$

h. No, two samples means will become similar, making the denominator close to zero. This causes the variance to explode.